Math 514, F19 **Name:** Answer

Quiz 5

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (4 points) Let $X(t)$, $t \geq 0$ be a Brownian motion process with drift and variance parameters $\mu = 2$ and $\sigma^2 = 16$, and $X(0) = 8$. Find $E[X(4)]$, $\mathrm{Var}(X(4))$ and $P(X(4) > 12)$.

$$E[X(4)] = E[X(0) + (X(4) - X(0))] = E[X(0)] + E[X(4) - X(0)]$$
$$= 8 + 2\cdot 4 = 16$$
$$\mathrm{Var}(X(4)) = \mathrm{Var}(X(0)) + \mathrm{Var}(X(4) - X(0)) = 0 + 16\cdot 4 = 64$$
$$\Rightarrow X(4) \sim N(16, 64)$$
$$P(X(4) > 12) = P\left(Z > \frac{\overbrace{12-16}^{=-4}}{\underbrace{\sqrt{64}}_{=8}}\right) = P(Z > -0.5) = P(Z < 0.5)$$
$$= \Phi(0.5) \approx 0.6915$$

Problem 2. (6 points) Let $S(t)$, $t \geq 0$ be a geometric Brownian motion process with drift parameter $\mu = 0.1$ and volatility parameter $\sigma = 0.4$. Find

(a) $P(S(3) < S(1) > S(0))$.

$X(t) \sim N(0.1 * t,\ 0.4^2 \times t)$, $X(0) = 0$

$$P(S(1) > S(0)) = P\left(\frac{S(1)}{S(0)} > 1\right) = P\left(\log\left(\frac{S(1)}{S(0)}\right) > 0\right) = P(X(1) > 0)$$
$$= P\left(Z > \frac{0 - 0.1 * 1}{0.4 * \sqrt{1}}\right) = P\left(Z > -\frac{1}{4}\right) = P(Z < 0.25) = \Phi(0.25) \approx 0.5987$$
$$P(S(3) < S(1)) = P\left(\frac{S(3)}{S(1)} < 1\right) = P\left(\log\left(\frac{S(3)}{S(1)}\right) < 0\right) = P(X(2) < 0)$$
$$= P\left(Z < \frac{0 - 0.1\times 2}{0.4\sqrt{2}}\right) = P\left(Z < -\frac{\sqrt{2}}{4}\right) = 1 - \Phi\left(\frac{\sqrt{2}}{4}\right) \approx 0.3618$$
$$\Rightarrow P(S(3) < S(1) > S(0)) = P(S(1) > S(0))\, P(S(3) < S(1))$$
$$\approx 0.5987 \times 0.3168 \approx 0.2166$$

(b) $E[S(3)]$ and $\mathrm{Var}(S(3))$ given $S(0) = 2$.

$S(3) = S(0)\, e^{X(3)} = 2e^{X(3)}$ where $X(3) \sim N(0.1\times 3,\ 0.4^2 \times 3) = N(0.3, 0.48)$

$$E[S(3)] = 2E[e^{X(3)}] = 2e^{E[X(3)] + \frac{1}{2}\mathrm{Var}(X(3))} = 2\cdot e^{0.3 + \frac{1}{2}\times 0.48} = 2e^{0.54} \approx 3.432$$
$$\mathrm{Var}(S(3)) = E[(S(3))^2] - (E[S(3)])^2 = E[4e^{2X(3)}] - (2e^{0.54})^2$$
$$= 4E[e^{2X(3)}] - 4e^{1.08} = 4\left(e^{E[2X(3)] + \frac{1}{2}\mathrm{Var}(2X(3))}\right) - 4e^{1.08}$$
$$= 4e^{0.6 + \frac{1}{2}\cdot 4\times 0.48} - 4e^{1.08} = 4(e^{1.56} - e^{1.08}) \approx 7.2566$$